

Explainable Prototype-Based Few-Shot Learning for Skin Lesion Classification

Ashish, Bhavya Ahuja, and Nikhil Kumar Rajput

Abstract—Deep learning has been highly successful in medical image analysis, but due to lack of availability of labeled data and lack of interpretability of such models, it has not yet been widely adopted in real-world clinical settings. Traditional convolutional neural networks (CNNs) are heavily dependent on large-scale annotated datasets which is scarce in medical and these models also work as opaque black-box models, so it makes them unsuitable for clinical applications and professionals where safety is top priority. In this paper, we present an explainable few-shot learning (XFSL) framework which integrates deep features learning, prototype-based classification, and similarity-based interpretability using the (Δ) metric, and this does not rely on computationally complex architectures or large-scale manual annotated data. The proposed framework is built upon the EfficientNet-B0 backbone, and then to learn domain-specific representations and also improving feature separability between visually similar classes (e.g., melanoma and benign nevi), it was fine-tuned and optimized with Focal Loss. Test-Time Augmentation (TTA) and Power Normalization were used to stabilize feature embeddings, which made the model more reliable from geometric perspective. Each query image was classified by measuring its similarity, by using cosine similarity. Beyond classical Grad-CAM visual explanations, we propose the Similarity-Gap (Δ) metric, for improved interpretability. This mathematically transparent metric measures the ambiguity in feature space between the top competing predictions and provides a more interpretable alternative to overconfident probability based scores. We use this metric to implement an automated out-of-distribution detection mechanism that detects and rejects query images if they fall below the threshold. The CNN backbone of proposed framework shows a very reliable performance by achieving a peak validation accuracy of 81.22% and an ROC-AUC of 0.9511 on multi-class dermoscopic dataset. Under extreme data constraints, an N-shot ablation study identifies 20-shot learning as the optimal baseline, yielding a Mean Accuracy of 76.64% and a Macro F1-score of 0.7634, indicating a clinically conservative prediction behavior. Furthermore, the $\tau = 0.55$ threshold OOD strategy demonstrates strong generalization, successfully rejecting 94.12% of non-medical images and 82.35% of unseen clinical lesions. Overall, the results indicate that interpretable and uncertainty-aware classification can be achieved under severe data constraints, representing a step toward more reliable and interpretable AI systems in clinical decision support.

Index Terms—Few-Shot Learning, Explainable AI, Medical Image Analysis, Deep Learning, Prototype-Based Learning, Grad-CAM, Computer Vision, Uncertainty Estimation

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I. INTRODUCTION

Recent advances in deep learning, particularly convolutional neural networks (CNNs), have significantly improved automated medical image analysis systems. These CNN models have shown strong performance in medical imaging tasks such as classification, detection, and segmentation tasks, in some cases they have produced expert-level detection [1], [2]. Even after such great achievements and success on many different levels, these models have not been widely adopted in real world medical settings because of many practical and complex methodological challenges.

The primary obstacle is the lack of large-scale labeled medical data. To get high quality annotations, it requires domain expertise which are costly in terms of person-hours, these are very time-consuming to generate, and also subject to strict privacy regulations. Due to this, many clinical datasets, especially for rare diseases or early-stage conditions, contain very few labeled samples per class [3]. In these cases, traditional deep learning models generally tend to overfit and have poor generalization performance.

The other big challenge is lack of interpretability. Most of the high performing deep learning models which have shown great results are black-box models with very little to almost no insight on how they reached to the final decision and their process involved in making that decision. In fields like healthcare, where safety is more important than anything else, this black-box nature of deep learning models reduce the trust of clinician and makes the regulatory approval even more strict and difficult [4], [5]. There is thus an increasing demand for models that are not only good in performance and accuracy but can also provide meaningful and reliable explanations.

To overcome this difficulty, Few-Shot Learning (FSL) has been developed as a promising approach for learning models that generalize from few examples helping address the data scarcity problem. In particular, prototype-based methods, where models represent each class as a centroid in an embedding space and then classify query images based on similarity measures [6], [7]. While these approaches work effectively in settings where data is scarce, but these approaches typically do not have mechanisms to estimate prediction uncertainty or offer any valuable interpretable reasoning.

At the same time, many different Explainable Artificial Intelligence (XAI) techniques are being developed to make models more transparent and interpretable, Grad-CAM is one such example which have been developed to improve model's transparency by highlighting regions of query image which significantly influence predictions [8]. However, these

are mostly post-hoc methods and these are not inherently integrated into the decision making process of model, and therefore it limits their ability to give reliable estimations on uncertainty.

In this paper, we propose an explainable few-shot learning (XFSL) framework for medical image classification which helps in bridging the gap between data-efficient learning and interpretable decision making system. This proposed framework integrates deep feature extraction with prototype based classification, which makes effective learning in the limited data scenario possible. An EfficientNet-B0 backbone [9] which is then optimized with Focal Loss [10], and fine-tuned to extract domain-specific features while also proactively tackling feature-space collision between visually-similar classes. To make this feature extraction more stable, test-time augmentation (TTA) and power normalization were used before creating prototypes to the framework, without any data reduction in sample. Then we classify the query samples by cosine similarity in the optimized embedding space.

To make the framework more interpretable, reliable, and trustworthy, we introduce a similarity-based uncertainty estimation mechanism which evaluates the distance between top competing class predictions. This formulation which is referred as the Similarity-Gap (Δ), directly measures the ambiguity of the feature-space and then allows model to distinguish between the predictions whether they are confident or uncertain predictions. This mathematically transparent approach captures the geometric structure of the learned embedding space with more faithfulness as compared to other such approaches like overconfident softmax probabilities [11].

We propose a threshold based rejection mechanism which is based on this metric with empirically obtained threshold ($\tau = 0.55$) to safely filter out-of-distribution (OOD) samples [12]. This approach allows the system to choose whether to make a prediction or abstain from making any forced predictions on unfamiliar or ambiguous inputs which improves the safety of clinical deployment.

We tested the proposed framework on a multi-class dermoscopic dataset and it shows that proposed framework provides interpretable predictions while maintaining stable performance under limited-data scenarios.

The framework combines data-efficient learning and embedding-space uncertainty estimation to provide a practical path towards making more trustworthy AI systems in healthcare. Crucially, the framework also demonstrates a clinically conservative diagnostic bias, where it prioritizes to recall malignant lesions rather than benign lesions for precision, aligns with the clinical safety margins. Unlike other traditional confidence estimation methods that require probability calibration, our proposed Similarity-Gap (Δ) works directly in the embedding space and hence it has a geometric interpretation of uncertainty without introducing extra model components or post-hoc calibration.

A. Contributions

The key contributions of this work are as follows:

- **Explainable Few-Shot Learning Framework:** We introduce a lightweight XFSL system which combines

similarity-based interpretability using the (Δ) metric with prototype-based classification. It offers a transparent decision-making pipeline that is independent from computationally demanding attention mechanisms or human spatial annotations.

- **Improved Feature Separability:** We show that during pre-training, use of Focal Loss successfully reduces the feature overlap between biologically similar lesion types (such as benign nevi and melanoma). This optimization technique gets a feature-space ROC-AUC of 0.9511, which shows that this optimization technique encourages distinct illness groupings.
- **Embedding Stabilization:** In order to compress extreme feature distributions and to improve 20-shot accuracy to 76.64% without requiring for aggressive outlier removal, we suggest combining Test-Time Augmentation (TTA) and Power Normalization.
- **Similarity-Gap (Δ) Metric:** Our suggestion is to quantify the geometric ambiguity between the top contending predictions using a feature-space uncertainty measure. This offers an alternative to overconfident softmax probabilities that is both theoretically straightforward and interpretable.
- **Clinically Conservative Diagnostic Bias:** We show that malignant recall is naturally prioritized in this framework over the benign precision, in the optimized 20-shot prototype space. Instead of forcing ambiguous lesions into false-negative benign classifications, this clinically conservative prediction tendency guarantees that they are treated with the proper clinical caution.
- **Uncertainty-Aware OOD Detection:** We develop an empirically validated rejection mechanism based on a cosine similarity threshold of $\tau = 0.55$. This mechanism demonstrates strong performance in filtering out-of-distribution samples, achieving up to 94.12% rejection of non-medical inputs and 82.35% rejection of unseen clinical lesions.

The remainder of this paper is organized as follows. Section II reviews related work in few-shot learning and explainable AI within the context of medical imaging. Section III describes the proposed explainable few-shot learning (XFSL) framework, including the Similarity-Gap (Δ) metric, out-of-distribution (OOD) detection mechanism, and Grad-CAM integration. Section IV presents the dataset, experimental setup, and evaluation metrics, while Section V presents the N-shot ablation studies and overall evaluation results. Section VI provides an in-depth analysis of the key findings, strengths, limitations, and potential future directions. Finally, Section VII concludes the paper.

II. RELATED WORK

A. Few-Shot Learning and Meta-Learning

Few-Shot Learning (FSL) aims to learn models capable of classifying novel categories using only a small number of examples. The primary objective is to learn discriminative feature representations that remain robust under extreme data

scarcity [6], [7]. A significant portion of existing literature relies on meta-learning, where models are trained across multiple tasks to learn generalizable distance metrics or initialization parameters. Prototypical Networks [6] use a mean embedding (prototype) to represent each class and they perform classification by using Euclidean distance, and therefore enforcing intra-class compactness. Learnable similarity functions and quick gradient-based adaptation methods are further introduced by variants like Relation Networks [13] and Model-Agnostic Meta-Learning (MAML) [3].

In recent times, research has been turned toward Explainable Few-Shot Learning (XFSL) to overcome the inherent opacity of these models. For example, Wang et al. [14] proposed MTUNet, which works by highlighting informative picture of regions by using combinations of pairwise matching with a self-attention-based pattern extractor. Although this method achieved 5-way 1-shot accuracies of 58.13% on mini-ImageNet [15], also the study suggested that visual patterns can facilitate information transfer. However, these methods frequently rely on computationally demanding self-attention modules. Additionally, while these approaches offer visual interpretability, but they lack integrated mathematical procedures to address diagnostic ambiguity or quantify feature-space uncertainty without appreciably adding architectural complexity.

B. Few-Shot Learning in Medical Imaging

Some of the major domain-specific difficulties in medical imaging classification process include substantial intra-class variability, with severe class imbalance, and a lack of labeled data [16]. Additionally, to do data annotation, specific specialist knowledge and adherence to strict privacy standards are needed, which leads to the scarcity and datasets with extremely limited sample sizes. In recent times, by using direct expert assistance, several studies have tried to increase the spatial realism of FSL. Using a Grad-CAM-based Dice loss [8], Uddin et al. [17] developed a framework that integrates radiologist-provided regions of interest (ROIs) during training, increasing the accuracy from 77.09% to 83.61% on the BraTS (MRI) dataset. Similar to this work, Wang et al. [18] presented EGxFSL, which achieves good performance on MRI and chest X-ray datasets by combining spatial supervision with active learning. Other than these direct spatial supervision works, Nikpour and Armanfard [19] introduced LaHA which is a reinforcement learning based approach that uses Vision-Language Models (VLMs) as a reward signals for models to identify the "hard attention" patches.

Although these methods have greatly increased the spatial fidelity, but at the same time they also create some practical bottlenecks: they either rely heavily on the expensive, manually annotated expert spatial masks [17], [18], or they require computationally expensive reinforcement learning agents and external VLMs [19]. This shows that the need of frameworks that can improve feature separability without the need of costly manual annotations or big architectural complexity.

C. Anomaly Detection and Reliability in Clinical AI

A very important requirement for the clinical adoption of AI systems is explainability [4], [20]. To provide safe deployment, models must be able to identify out-of-distribution (OOD) inputs from query in addition to being interpretable for safe use. Deviation Networks (DevNet), which learns end-to-end differentiable anomaly scores by enforcing statistically significant deviations for anomalous data, and these methods were used by Pang et al. [12] to address these issues of anomaly detection. Nevertheless, DevNet makes extensive use of complicated Multiple Instance Learning (MIL) formulas and artificial Gaussian priors.

Additionally, some other popular explainability techniques like Grad-CAM [8] and LIME [21] gives another approach in addition which offers qualitative spatial insights but they do not intrinsically quantify mathematical ambiguity between competing multi-class predictions, as noted by Lee et al. [22]. This gap emphasizes the need of a lightweight, intrinsic safety mechanism which does not have to rely solely on artificial data distributions in order to construct a trustworthy OOD rejection criteria directly within the embedding space and quantify feature-space ambiguity mathematically.

D. Identified Gaps and Context of This Work

A review of the literature shows that existing XFSL approaches either rely on costly expert ROI masks [17], [18] or computationally intensive reinforcement learning agents [19]. This work addresses three key gaps:

- 1) **Improved Feature Separability:** Prior medical FSL studies often rely on standard loss functions that fail to adequately separate visually similar clinical classes. We incorporate Focal Loss during pre-training to explicitly reduce feature overlap between Melanoma and Benign Nevi without requiring expert masks. Furthermore, we integrate Test-Time Augmentation (TTA) and Power Normalization to inherently stabilize these embeddings, eliminating the need for aggressive data-dropping heuristics.
- 2) **The Similarity-Gap (Δ):** Existing models rely heavily on overconfident softmax probabilities [4], [23]. We introduce a mathematically transparent feature-space Δ -metric to quantify diagnostic ambiguity between top competing class prototypes.
- 3) **Lightweight OOD Safety:** We introduce an empirically validated cosine-similarity thresholding mechanism ($\tau = 0.55$) that successfully rejects 94.12% of random non-medical images. Unlike complex binary anomaly detectors [12], this provides an inherent, computationally light safety mechanism for multi-class medical systems.

This proposed approach offers a more practical clinical decision-support framework by shifting the focus from purely visual heatmap explanations to quantifiable feature-space safety.

III. METHODOLOGY

The proposed framework integrates deep feature extraction, prototype-based Few-Shot Learning (FSL), and multi-layered

interpretability into a unified medical image classification pipeline. The overall process comprises four key stages: (1) feature extraction using a convolutional neural network, (2) prototype construction from a restricted support set, (3) similarity-based classification of query samples, and (4) the generation of visual and reasoning-based explanations.

By coupling robust representation learning with similarity-based inference, the framework achieves effective learning under limited data conditions while also maintaining transparency at the decision level. The overall architecture of the proposed system is illustrated in Fig. 1.

A. Feature Extraction using CNN

Feature extraction is performed using an EfficientNet-B0 [9] architecture, which provides an optimal balance between computational efficiency and strong representational capacity. This choice aligns with the overall objective of maintaining a lightweight framework suitable for real-world clinical deployment. The network is fine-tuned on the pre-training dataset to adapt to the domain-specific morphological characteristics of dermoscopic lesions.

After optimization, the final classification layer is removed, and the network is repurposed as a dedicated feature extractor. To ensure geometric stability against morphological variations, the framework applies Test-Time Augmentation (TTA) by passing the original, horizontally flipped, and vertically flipped images through the backbone and averaging their outputs. Power Normalization (Tukey's Transform) is then applied to further compress extreme feature-space values by taking the signed square root of the averaged vector. This results in a stable feature embedding $f(x) \in \mathbb{R}^d$, where the embedding dimension is $d = 1280$. These dense vectors serve as the foundation for prototype construction and similarity-based inference.

B. Prototype-Based Few-Shot Learning

Prototype-based Few-Shot Learning (FSL), inspired by Prototypical Networks [6], represents each class as a central embedding. In the proposed framework, each class is represented by a single prototype defined from a small support set. Let $\mathcal{S}_c = \{x_i^c\}_{i=1}^K$ denote the support samples for class c , where K represents the shot size.

Since the CNN backbone is optimized using Focal Loss and the extracted features are further stabilized using TTA and Power Normalization, aggressive outlier filtering becomes unnecessary. Therefore, the prototype p_c is computed as the mean of all embeddings in the support set:

$$p_c = \frac{1}{K} \sum_{x_i \in \mathcal{S}_c} f(x_i) \quad (1)$$

The resulting prototype p_c and all query embeddings $f(x_q)$ are strictly L2-normalized to ensure stability and geometric consistency during cosine similarity computation.

C. Similarity-Based Classification

During inference, the feature embedding $f(x_q)$ of a query image x_q is extracted and compared against all class prototypes. Since both the query embeddings and the prototypes are L2-normalized, cosine similarity simplifies to the dot product:

$$S(x_q, p_c) = f(x_q) \cdot p_c \quad (2)$$

The predicted class $\hat{y}$ is assigned to the class with the highest similarity score:

$$\hat{y} = \arg \max_c S(x_q, p_c) \quad (3)$$

In addition to the primary prediction, the framework retains the second-highest similarity score. This enables quantitative analysis of prediction ambiguity and forms the basis for the uncertainty estimation mechanism described next.

D. Similarity-Gap (Δ) Ambiguity Quantification

Softmax probabilities [11] are frequently used by conventional neural networks to estimate confidence. But for most of the times, these methods are overconfident, giving false forecasts a high degree of certainty, which is especially troublesome in unclear medical situations.

We provide a similarity-based uncertainty measure that functions directly in the feature embedding space in order to overcome this constraint. Cosine similarity is calculated by using all class prototypes for a query sample. The greatest and second-highest similarity scores of samples are denoted by $S_{\text{top-1}}$ and $S_{\text{top-2}}$, respectively. The definition of the Similarity-Gap (Δ) is:

$$\Delta = S_{\text{top-1}} - S_{\text{top-2}} \quad (4)$$

The size of Δ is a direct measure of the reliability of prediction. A large Δ indicates high confidence, i.e. the predicted class is well separated from competing classes. On the other hand, a smaller Δ indicates an overlap in the embedding space, which corresponds to high diagnostic ambiguity.

In contrast to probability-based confidence scores, Δ is an explicit measure of feature-space separability, which is well aligned with learned representations. This yields a transparent and interpretable measure of uncertainty in prediction, which is important for reliable clinical decision making.

E. Unknown Detection (OOD Rejection)

To further improve system reliability, an uncertainty-aware rejection mechanism is incorporated into the decision layer. If the maximum similarity score for a query sample falls below a predefined threshold τ , the sample is classified as an out-of-distribution (OOD) [12] input. The prediction rule is defined as:

$$\hat{y} = \begin{cases} \text{UNKNOWN}, & \text{if } \max_{c \in \mathcal{C}} S(x_q, p_c) < \tau \\ \arg \max_{c \in \mathcal{C}} S(x_q, p_c), & \text{otherwise} \end{cases} \quad (5)$$

where $\mathcal{C}$ denotes the set of all classes.

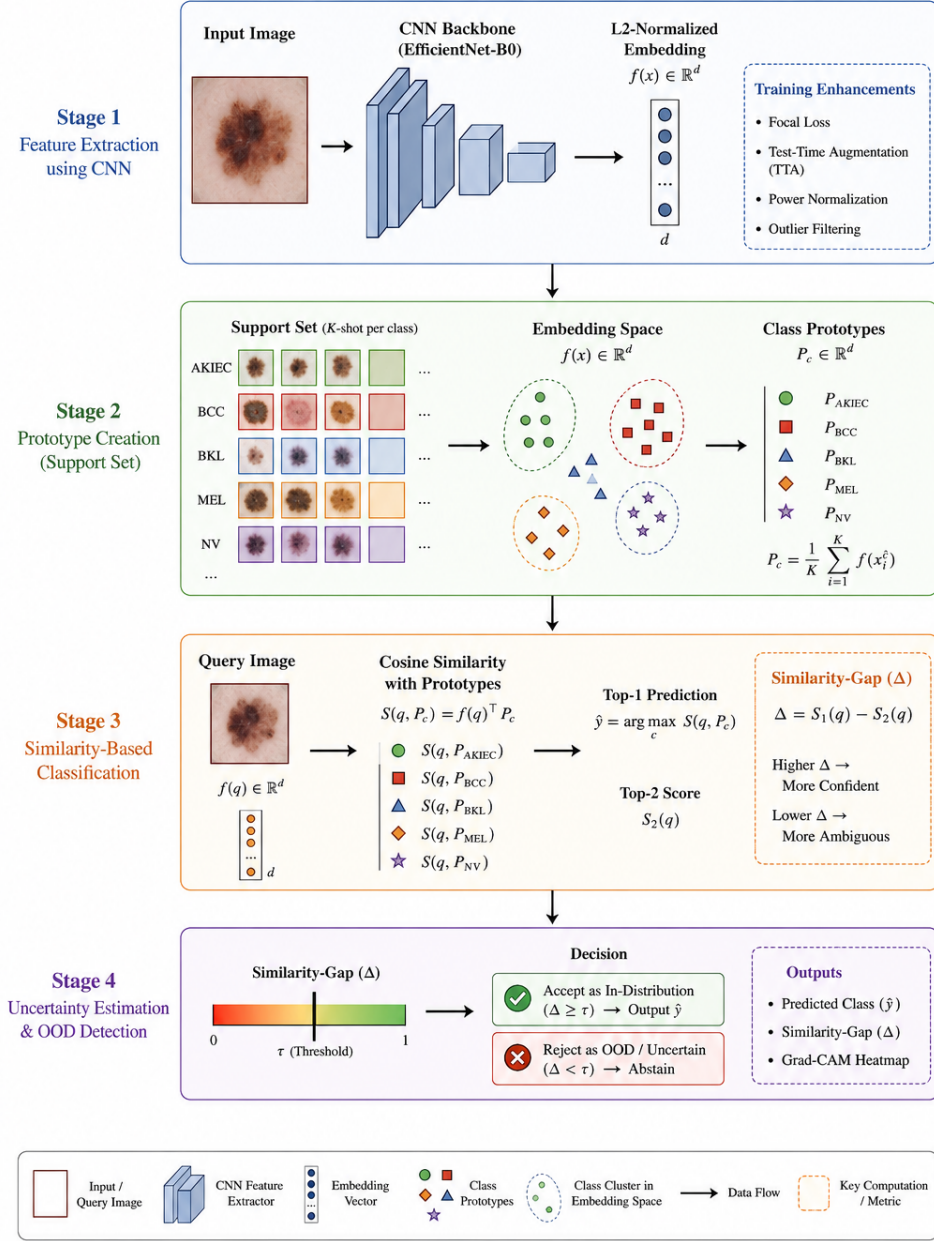


Fig. 1. Overview of the proposed XFSL framework for medical image classification. This pipeline consists of four main stages: (1) CNN-based feature extraction which was enhanced during training, (2) prototype creation from the support set for all classes, (3) similarity-based classification using cosine similarity and the Similarity-Gap (Δ) metric, and (4) uncertainty estimation with out-of-distribution (OOD) detection. This proposed framework integrates deep representation learning with interpretable and uncertainty-aware decision making.

The threshold τ is empirically determined to balance in-distribution accuracy and anomaly rejection. In this framework, τ is set to 0.55. This mechanism acts as an automated safety layer, preventing unreliable predictions on unfamiliar inputs and improving robustness in real-world clinical deployment.

F. Explainability using Grad-CAM

Grad-CAM is used to create qualitative visual explanations in addition to quantitative uncertainty estimation provided by the Δ metric [8]. Grad-CAM calculates class-specific

activation maps by taking gradients of class-scores in the activation map corresponding to the last convolutional layer.

Localization maps generated with these results are upsampled and overlaid onto the input image to show the regions with the strongest morphology that affect the prediction of the model (e.g., pigment networks, irregular borders). Further smoothing and blendings are used for visual clarity.

Both Grad-CAM and Δ contribute to the overall understandability of the decision-making process in this proposed dual-layer explainability framework, while the Grad-CAM feature provides spatial information about the model's focus on the

image, Δ mathematically quantifies the prediction ambiguity of model.

G. Inference Pipeline

Inference is the process where a query image is passed into the feature extractor which then returns the $L2$ -normalized image embedding for the query. These embeddings are then compared to all class prototypes present based on cosine similarity, and then it produces a ranked list of candidate diagnoses.

The final system output has three main elements: (1) the class label predicted using the maximum similarity, (2) the Similarity-Gap (Δ) value which reflects the reliability of prediction and (3) a heatmap of the clinically relevant spatial regions, which is obtained using Grad-CAM. Importantly, when the maximum similarity score is below the threshold $\tau = 0.55$ it is a safe interception and the input is classified as *UNKNOWN*.

This multi-layered output can be augmented with qualitative spatial validation, and thus it yields a transparent, reliable, and interpretable decision Support framework for clinical application.

IV. EMPIRICAL VALIDATION

In this section, we describe the dataset, experimental setup, and evaluation metrics used in this study.

A. Dataset and Preprocessing

Experiments are conducted on a subset of the HAM10000 dataset [16], available through the ISIC (International Skin Imaging Collaboration) archive, which serves as a widely used benchmark for skin lesion analysis. Five clinically relevant classes are evaluated: AKIEC (Actinic Keratoses and Intraepithelial Carcinoma), BCC (Basal Cell Carcinoma), BKL (Benign Keratosis), NV (Melanocytic Nevus), and MEL (Melanoma). The choice of these classes was made because they were chosen for being of diagnostic importance and having a substantial amount of visual overlap, making this a stringent testbed for class performance and interpretability.

The dataset is split into two sets: one large pre-training set for CNN feature extraction, and a strictly balanced query set for the FSL evaluation. During CNN pre-training, the whole ISIC dataset is used without modifying the natural class imbalance from clinical practice which ranges from around 700 images with the rare diseases (e.g. AKIEC) to more than 12,000 images with common benign diseases (NV). This imbalance leads us to attempt to address the problem of the majority-class bias in representation learning by using Focal Loss [10] as a method.

The episodic FSL evaluation set is, on the other hand, made up of exactly 151 unseen images per class. Given the optimal value of $K = 20$, 20 images are randomly selected from the set as the support set with the remaining 131 images are used as the query set.

All images are resized to 224×224 pixels and normalized using ImageNet [15] mean and standard deviation values. To

improve generalization and reduce overfitting, a robust data augmentation strategy is applied, including random 360-degree rotations, affine transformations (scaling and shearing), and color jittering. These augmentations encourage the model to learn orientation-invariant morphological features rather than relying on superficial artifacts.

B. Experimental Setup

The proposed model is evaluated on a five-class dermoscopic image dataset. The data is divided into a heavily imbalanced training set used exclusively to fine-tune the CNN backbone, and a perfectly balanced test set (151 images per class) used for few-shot evaluation. The EfficientNet-B0 [9] CNN backbone is fine-tuned for 8 epochs using the Adam optimizer. Due to the inherent difficulty of separating visually similar classes, such as Melanoma (MEL) and Benign Nevi (NV), Focal Loss [10] is utilized instead of standard cross-entropy. Focal Loss dynamically down-weights easy samples and focuses on hard-to-classify examples, thereby improving feature separation in overlapping regions of the embedding space.

In the Few-Shot Learning (FSL) evaluation, episodic support-query partitions are constructed. Specifically, K images per class are randomly sampled to construct class prototypes, and the remaining images are used as query samples. This process is repeated over 10 independent runs with random sampling to ensure statistical robustness.

C. Evaluation Metrics

The performance of the framework is evaluated using the following metrics:

- Training and Validation Accuracy
- Macro F1-Score and Matthews Correlation Coefficient (MCC)
- ROC-AUC (Embedding Space Separability)
- N-Shot Mean Accuracy and Standard Deviation
- Out-of-Distribution (OOD) Detection Rate

V. RESULTS

A. CNN Training Performance

The CNN backbone performs well, as seen in Fig. 2 and summarized in Table I.

TABLE I
CNN TRAINING PERFORMANCE (EPOCH 8)

Metric	Value
Final Training Accuracy	83.23%
Best Validation Accuracy	81.22%
Validation ROC-AUC	0.9511
Macro F1-Score	0.7358

With a final training accuracy of 83.23% and a peak validation accuracy of 81.22%, the model shows little overfitting. The ROC-AUC notably hits 0.9511, indicating that Focal Loss efficiently divides the feature embedding space into distinct illness clusters. This strong feature representation serves as the basis for the few-shot evaluation that follows.

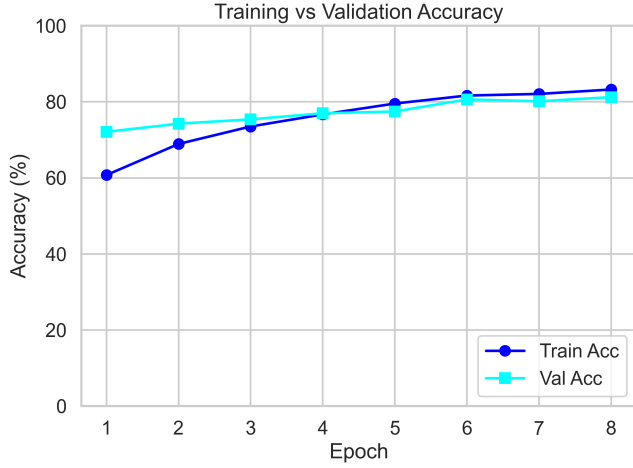


Fig. 2. Training and validation accuracy over 8 epochs show close alignment, indicating effective mitigation of overfitting.

B. XFSL Performance and N-Shot Ablation

To evaluate performance under constrained data conditions, an N-shot ablation study is conducted by varying the support set size K from 5 to 30. As shown in Fig. 3, performance improves rapidly from $K = 5$ to $K = 15$, followed by a clear stabilization phase.

The best baseline of the framework is achieved at $K = 20$ with mean accuracy of 76.64%. It is also worth noting that the Macro F1-score (0.7634) is quite close to the accuracy, while the MCC reaches 0.7105. These results of experiment shows indication that all classes are predicted with similar accuracy and hence confirm the effectiveness of the combination of Focal Loss and embedding stabilization to reduce the bias of the majority class. Moreover, the performance robustness is shown by a small standard deviation of 1.23% over 10 runs.

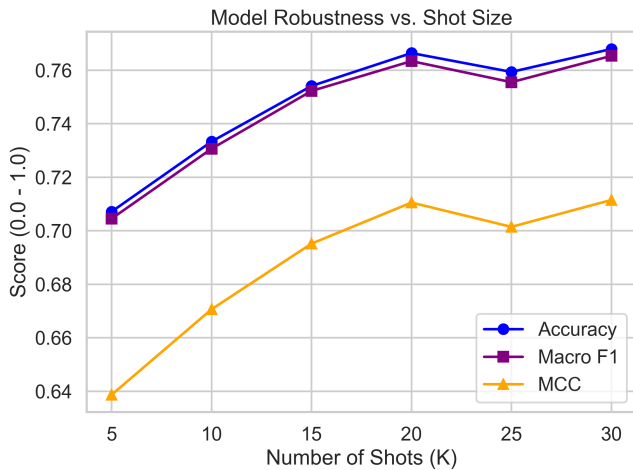


Fig. 3. Model robustness across varying support set sizes (K). Performance stabilizes at $K = 20$, establishing the optimal baseline.

When going beyond $K = 20$ we see diminishing returns (e.g. 75.94% for $K = 25$), which indicates that additional

TABLE II
FEW-SHOT LEARNING PERFORMANCE AT OPTIMAL $K = 20$

Metric	Value
Mean Accuracy	76.64%
Macro F1-Score	0.7634
Matthews Correlation Coefficient (MCC)	0.7105
Standard Deviation	1.23%
Number of Runs	10

samples increase variability and noise in the prototype representations rather than enhancing them. And therefore the optimal support size for this dataset was observed at $K = 20$.

C. Per-Class Accuracy and Clinical Bias

Table III presents per-class performance at the optimal 20-shot setting. The results, supported by Fig. 4, highlight varying performance across morphological classes.

TABLE III
PER-CLASS PERFORMANCE AT OPTIMAL $K = 20$

Class	Precision	Recall	F1-Score
AKIEC	0.81	0.93	0.87
BCC	0.84	0.81	0.82
BKL	0.80	0.75	0.77
MEL	0.62	0.76	0.69
NV	0.87	0.64	0.74

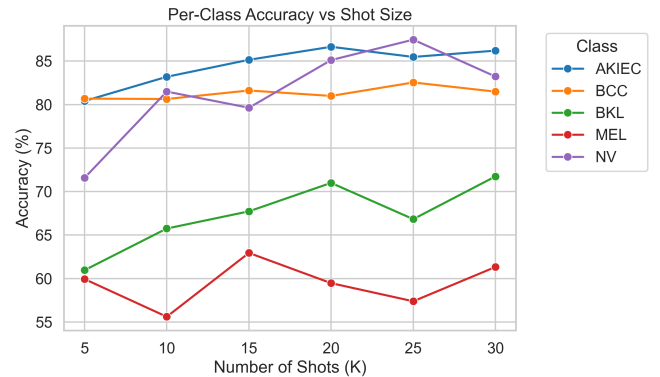


Fig. 4. Per-class accuracy across different N-shot support sizes. Melanoma (MEL) stabilizes near $K = 20$ due to feature-space optimization.

The model achieves high sensitivity for distinct classes such as AKIEC, with a recall of 93%. A key clinical observation is the model's behavior for Melanoma (MEL) and Benign Nevi (NV). While NV shows high precision (87%), its lower recall (64%), combined with a Melanoma recall of 76%, indicates a conservative diagnostic bias. The model prioritizes minimizing false negatives in malignant cases by cautiously classifying ambiguous benign lesions as potentially malignant.

D. Baseline Comparison

In order to compare the capabilities of the proposed XFSL framework with five existing classification baselines and

understand its effectiveness after comparing with standard metrics, the performance of the same was compared under the same 20-shot setting. They encompass metric-learning methods (Vanilla Prototypical Networks, Matching Networks and k-Nearest Neighbors) and conventional machine learning classifiers (Random Forest and Linear Support Vector Machine) while all using the same EfficientNet-B0 embeddings.

As demonstrated in Table IV the proposed XFSL framework is competitive in its performance with other baseline methods, while simultaneously providing interpretability and uncertainty estimation. It attains a mean accuracy of 76.64% and a Macro F1-score of 0.7634. Existing methods based on a metric point of view like k-NN (71.63%) or Matching Networks (72.92%) show comparatively lower performance in the high dimensional embedding space, and Vanilla ProtoNet (75.89%) offers a good baseline. Test-Time Augmentation (TTA) is combined with Power Normalization to bring simple metric learning closer to more sophisticated classification methods.

TABLE IV
PERFORMANCE COMPARISON AGAINST BASELINES AT $K = 20$

Model	Mean Acc (%)	Macro F1	MCC
k-Nearest Neighbors ($k = 3$)	71.63	0.7103	0.6484
Matching Networks	72.92	0.7209	0.6694
Random Forest	75.07	0.7464	0.6901
Vanilla ProtoNet	75.89	0.7551	0.7008
Linear SVM	77.08	0.7667	0.7157
Proposed XFSL Framework	76.64	0.7634	0.7105

A Linear SVM works as a rigid classifier despite achieving a little greater raw accuracy (77.08%). In particular, SVM creates decision boundaries in the form of hyperplanes, which then requires a prediction for each input without a built-in rejection mechanism for query. This trend may lead to over-confident predictions on out-of-distribution (OOD) samples, which can be dangerous in clinical contexts.

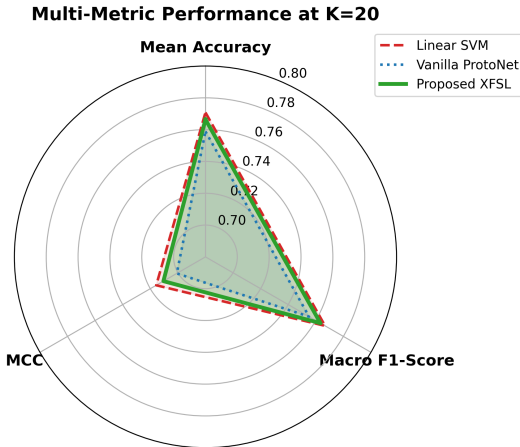


Fig. 5. Radar chart illustrating the multi-metric performance trade-off at $K = 20$. The proposed XFSL framework demonstrates a balanced geometric profile across Accuracy, Macro F1, and MCC, bridging the gap between simple metric learning (Vanilla ProtoNet) and rigid classifiers (Linear SVM).

On the other hand, in order to preserve well-bounded

geometric prototypes, which are the suggested XFSL framework purposefully forfeits a little 0.44% decrease in raw accuracy. This design decision permits the use of a safety threshold ($\tau = 0.55$) and also allows the computation of the Similarity-Gap (Δ) which is one of important part of this framework. As a result, the framework is more appropriate for practical medical use, where accurate uncertainty estimation is essential. Fig. 5 provides visual support for this multi-metric performance profile, demonstrating the framework's balanced stability across all evaluation criteria.

E. Feature Space Visualization (t-SNE)

To qualitatively validate the discriminative capability of the fine-tuned CNN backbone, t-Distributed Stochastic Neighbor Embedding (t-SNE) [24] is used to project the 1280-dimensional query set embeddings into a 2D space (see Fig. 6).

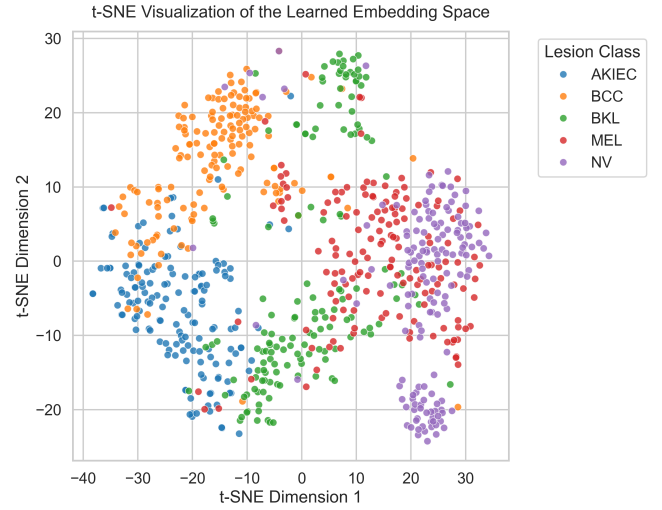


Fig. 6. t-SNE projection of the test set embeddings. The visualization shows strong geometric separation for distinct classes (e.g., AKIEC and BCC), along with expected biological overlap between Melanoma (MEL) and Benign Nevi (NV).

As per projection, the model maps dermatological morphological features to separate geometric clusters. For classes with clear separation, such as AKIEC and BCC, the margin between the classes is quite high. Melanoma (MEL) and Benign Nevi (NV) are, however, near each other and exhibit a certain overlap in the embedding space. This observation is in accordance with known clinical features of dermoscopy, which indicates that the uncertainty of the model is not the result of a method limitation.

F. Confusion Matrix Analysis

To further analyze classification behavior, a pooled confusion matrix for the optimal baseline evaluation is generated. As shown in Fig. 7, most misclassifications occur between classes with high morphological similarity.

In particular, the confusion matrix reinforces the previously observed conservative bias. A significant portion of true NV (Benign Nevi) samples are classified as MEL (Melanoma).

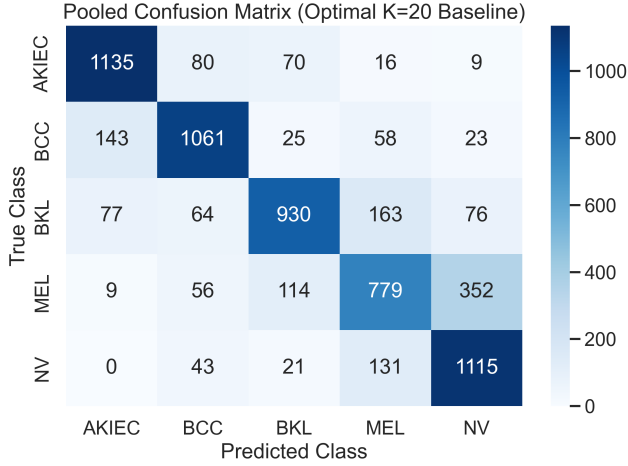


Fig. 7. Aggregate confusion matrix at $K = 20$. Misclassifications exhibit a conservative bias, prioritizing Melanoma (MEL) recall over benign precision.

This likely reflects clinical ambiguity rather than purely random error, indicating that the model captures meaningful but overlapping feature representations and defaults to the clinically safer class when uncertain.

G. Confidence and Similarity Gap Analysis

Similarity-based measures are analyzed to quantitatively assess prediction reliability.

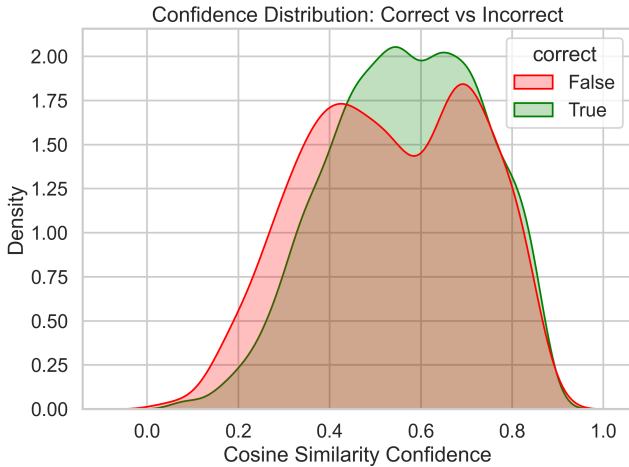


Fig. 8. Distribution of cosine similarity confidence scores for correct and incorrect predictions. Correct predictions generally exhibit higher similarity confidence values, while the incorrect predictions are mostly concentrated towards the lower confidence regions, although there remains a noticeable overlap between the two distributions.

As shown in Fig. 8, that while incorrect predictions are more concentrated in lower-confidence regions, correct predictions generally show greater cosine similarity confidence values. Nonetheless, there is still a discernible overlap between these two distributions, which suggests that top-1 similarity confidence might not always be enough to distinguish between confident and ambiguous predictions.

The Similarity-Gap (Δ) metric is used as a supplementary uncertainty estimation measure because of this observation. The distribution of the Similarity-Gap (Δ) between the top-1 and top-2 predictions is plotted, separating correct and incorrect predictions (see Fig. 9).

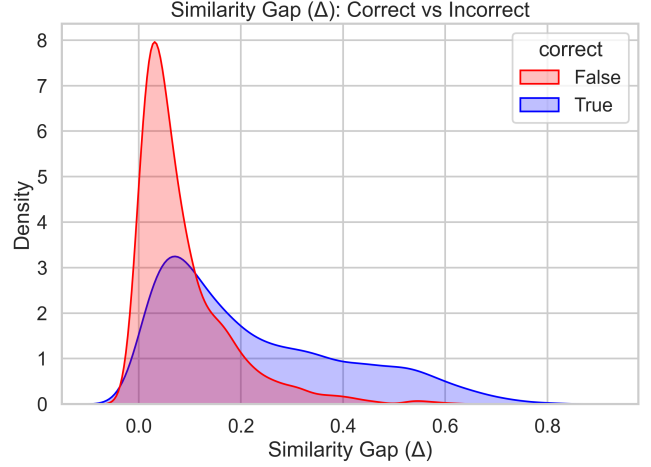


Fig. 9. Distribution of the Similarity-Gap (Δ). Incorrect predictions are strongly concentrated near 0.0, validating Δ as a reliable indicator of model uncertainty.

The resulting distributions support the effectiveness of the Δ metric. Incorrect predictions are heavily skewed toward 0.0, indicating high feature-space ambiguity. In contrast, correct predictions show a wider distribution, corresponding to greater confidence and stronger geometric separation. This demonstrates that Δ serves as a transparent and reliable indicator of prediction uncertainty, providing a more interpretable alternative to overconfident softmax probabilities [11].

H. Empirical Unknown (Out-of-Distribution) Detection

To evaluate robustness and clinical safety, the framework is tested on Out-of-Distribution (OOD) [12] samples across multiple threshold values. The evaluation dataset includes 51 Far-OOD images (random non-medical images) and 51 Near-OOD images (unseen clinical lesion types such as Dermatofibroma, Squamous Cell Carcinoma, and vascular lesions).

TABLE V
OOD DETECTION RATES ACROSS THRESHOLDS (τ)

Threshold (τ)	Far-OOD (%)	Near-OOD (%)
0.50	86.27%	68.63%
0.55	94.12%	82.35%
0.60	98.04%	96.08%

The framework demonstrates strong safety boundaries, as detailed in Table V and visualized in Fig. 10. At the empirically selected threshold of $\tau = 0.55$, the model achieves a rejection rate of 94.12% of Far-OOD samples and 82.35% of dangerous Near-OOD medical samples. A sensitivity analysis reveals that lower thresholds (e.g., $\tau = 0.50$) allow too many unseen samples to pass, while higher thresholds (e.g.,

$\tau = 0.60$) theoretically reject 98.04% of non-medical OOD samples but drastically increase the risk of incorrectly rejecting valid in-distribution images. Therefore, $\tau = 0.55$ is confirmed as the optimal trade-off between clinical safety and diagnostic usability.

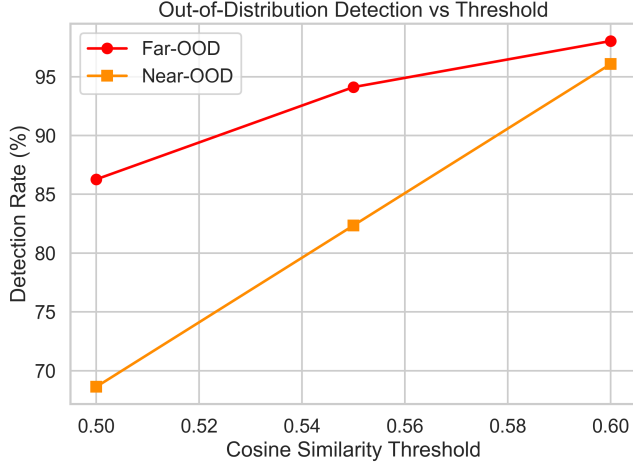


Fig. 10. Out-of-Distribution detection rates across varying cosine similarity thresholds. The $\tau = 0.55$ threshold balances optimal safety with clinical usability.

I. Qualitative Results and Explainability

Qualitative analysis was performed using Grad-CAM visualizations to interpret the spatial focus of the model.

The generated heatmaps (Fig. 11) confirm that the model accurately focuses on clinically relevant regions of the lesion—such as irregular borders and pigment networks—substantiating the validity of the learned representations. When combined with the quantitative Δ metric and the *UNKNOWN* detection threshold, these visual explanations provide a comprehensive, multi-layered insight into both the model’s spatial focus and its underlying reasoning confidence.

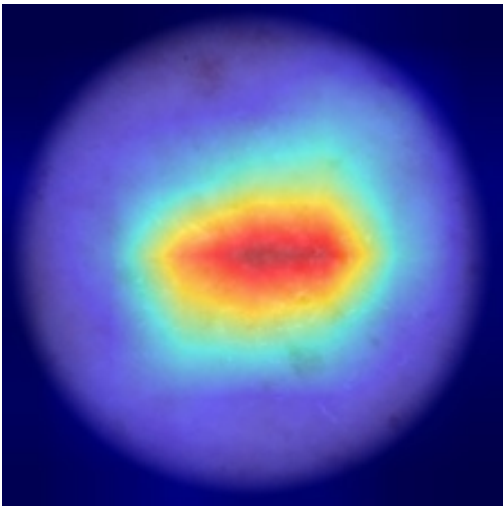


Fig. 11. Grad-CAM visualization highlighting the regions of the lesion that most strongly influenced the model’s similarity prediction.

J. Summary of Findings

The experimental results highlight the following key observations:

- The CNN backbone provides well-separated feature representations, achieving an ROC-AUC of 0.9511.
- The prototype-based FSL approach maintains a stable 76.64% accuracy using only 20 samples, effectively improving feature separability between majority-class bias (Macro F1-score of 0.7634).
- Per-class analysis reveals a conservative, clinically conservative prediction tendency that prioritizes Melanoma recall over benign precision.
- The Similarity-Gap (Δ) metric provides a mathematically grounded and interpretable measure of prediction uncertainty.
- The empirically validated threshold $\tau = 0.55$ acts as an effective safety mechanism, rejecting up to 94.12% of unfamiliar inputs.

Overall, the framework successfully balances data efficiency, clinical safety, and interpretability.

VI. DISCUSSION

A. Analysis of Results

The experimental findings show that the suggested framework has successfully adapted to a compromise between clinical interpretability, data efficiency, and accuracy. Even with a large class imbalance, the CNN backbone successfully learns discriminative feature representations for different classes with a peak validation accuracy of 81.22% and a ROC-AUC of 0.9511.

In addition to this, the few-shot learning part has emphasized the robustness of the proposed approach. The average accuracy of the model stands at 76.64% which is at the optimal 20-shot configuration with a standard deviation of 1.23%. In addition, other metrics like Macro F1-score (0.7634) and MCC (0.7105) values depict balance between the classes. This is a confirmation that the combination of Focal Loss and embedding stabilization (by using TTA and Power Normalization) is a successful approach for minimizing the majority class bias.

Per-class analysis shows clinically relevant behavior in difficult cases. While classes such as AKIEC do well (93% recall), other classes such as Melanoma (MEL) and Benign Nevi (NV) have significant feature-space overlap. The model conservatively addresses this ambiguity, with a strong 76% recall for Melanoma at the expense of some benign precision. This behavior suggests that the embedding space tends to reduce false negatives among malignant cases, which is in line with clinical safety considerations.

B. Effectiveness of the Proposed Approach

In low-data scenarios, the combination of prototype-based categorization with deep feature extraction works very well. While Test-Time Augmentation (TTA) and Power Normalization guarantee consistent and universal performance without need of huge dataset, balanced datasets or aggressive outlier

filtering, the CNN backbone creates discriminative embeddings.

Importantly, the Similarity-Gap (Δ) metric gives a clear approach for uncertainty estimation. In contrast to softmax probabilities, Δ is an actual measurement of the geometric distance in the embedding space between the competing class representations. The empirical distribution of Δ shows that the majority of the wrong predictions occur close to zero and this allows the use of a criterion to identify ambiguous cases.

The uncertainty-aware rejection mechanism of model contributes further to the system safety. The model classifies low-similarity samples (i.e., $\max S(x_q, p_c) < \tau = 0.55$) as *UNKNOWN* with a rejection rate of 94.12% for Far-OOD samples and 82.35% for unseen clinical lesions, and thus it avoids forced predictions on unfamiliar inputs.

The principal strengths of the proposed framework are summarized as follows:

- **Data Efficiency:** It demonstrates a competitive performance under low data constraints while using prototype-based Few-Shot Learning.
- **Class Balancing and Stability:** The integration of Focal Loss with Test-Time Augmentation (TTA) and Power Normalization has effectively mitigated the class imbalance and also morphological noise without removing data or filtering, achieving a robust Macro F1-score of 0.7634.
- **Uncertainty Estimation:** The Similarity-Gap (Δ) metric provides a mathematically grounded, feature-space-aligned measure of predictive reliability.
- **Robustness:** The threshold mechanism ($\tau = 0.55$) acts as a strict and automated safety boundary against out-of-distribution inputs.
- **Interpretability:** The synergistic combination of Grad-CAM with the Δ metric provides a cohesive visual and quantitative explanations for the query.

C. Comparison with State-of-the-Art Approaches

The suggested methodology shows that competitive performance can be obtained with a fraction of the annotated data as compared to traditional supervised CNN models. With just 20 support images per class, the 20-shot framework maintained 76.64% accuracy compared to the fully supervised baseline's 81.22% accuracy. Fully supervised models continue to be inflexible, opaque black-boxes even if they perform somewhat better in data-rich circumstances.

More importantly, our approach has successfully overcome the major limitations of recent Explainable Few-Shot Learning (XFSL) frameworks. Recent medical imaging models achieve high spatial fidelity, but either they rely heavily on costly manually annotated expert ROIs [17], [18], or they require computationally intensive reinforcement learning agents [19]. Similarly, state-of-the-art Out-of-Distribution (OOD) detectors usually rely on complex synthetic prior generation [12].

In contrast, we gain separable disease clusters (AUC 0.9511), quantifiable uncertainty (Δ), and robust OOD rejection, just by means of embedding space mathematics and feature stabilization. It is an effective and scalable solution for medical applications where there are multiple competing problems to solve, including data scarcity and interpretation.

D. Limitations

The suggested framework has some drawbacks even after its advantages that call for more research and detailed evaluation. Firstly, the prototype-based formulation makes the assumption that a single mean centroid in the embedding space may represent each class. Classes with substantial intra-class variability may be oversimplified by this assumption and such classes can have more variation and might need more prototypes, and therefore a multi-modal or distribution-based representations might more accurately portray the richness of the data.

Second, an out-of-distribution detection threshold $\tau = 0.55$ is set empirically for this dataset, it was obtained with testing other thresholds (e.g., 0.5 and 0.6) but it works on this dataset and may not adapt with this empirical setting on other datasets. The framework may need to be re-evaluated for other threshold or any automated detected threshold in other clinical databases or imaging modalities. Last, while Grad-CAM is valuable for giving spatial explanations, it is not necessarily clinically valid. The marked areas need to be checked against expert marked ground truth data before it can be used in the real world.

The evaluation is conducted on a subset of the HAM10000 dataset, which, although is widely used, but it represents a single imaging modality and clinical distribution. Therefore, the generalizability of the framework across other different clinical settings and imaging modalities (e.g., histopathology, radiology) remains to be tested.

E. Future Work

Future work will focus on improving both predictive performance and clinical interpretability. Potential directions include:

- Developing a multi-prototype or distribution-based representations to better capture intra-class variability which might need more support size for other prototypes.
- Incorporating a localized attention mechanisms to improve fine-grained feature discrimination in highly overlapping classes.
- Conducting quantitative evaluations with the help of expert-annotated bounding boxes to validate the clinical relevance of Grad-CAM heatmaps.
- Exploring adaptive and dynamically computed thresholding strategies for different datasets, cameras, and images to optimize out-of-distribution detection across different clinical environments.

VII. CONCLUSION

In this paper, we introduce an explainable few-shot learning (XFSL) approach to tackle the two most important issues in medical image diagnosis when trying to adopt AI for this which are, restricted interpretability of deep learning models and data scarcity. The suggested method exhibits significant robustness under limited data conditions by combining geometric prototype-based classification with EfficientNet-B0 feature extraction. The framework successfully reduces morphological variances at the embedding level by utilizing Power Normalization and Test-Time Augmentation (TTA),

which lessens the requirement for aggressive data-filtering techniques.

In addition, we propose the Similarity-Gap (Δ) metric, a metric for prediction uncertainty in the embedding space, based on geometric considerations. This formulation provides an interpretable basis for reasoning about diagnostic ambiguity, without the overconfidence problems of probability-based scoring methods.

Experimental results demonstrate that the proposed architecture achieves competitive performance among few-shot approaches in a 20-shot setting, with a mean accuracy of 76.64%. Importantly, the model deliberately sacrifices a small amount of raw accuracy compared to rigid linear classifiers (e.g., SVMs) to preserve bounded geometric prototypes. This design enables the use of a similarity-based safety threshold ($\tau = 0.55$) for detecting and rejecting out-of-distribution (OOD) inputs. Additionally, the learned prototype space exhibits a conservative bias that prioritizes the recall of malignant lesions, aligning with clinical safety requirements.

In conclusion, this study demonstrates that combining data-efficient representation learning, embedding stabilization, and embedding-space uncertainty estimation results in a practical and interpretable framework for medical decision support. Future work will explore multi-prototype representations to better capture intra-class variability and investigate adaptive uncertainty thresholds across larger, multi-modal clinical datasets.

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